

[[user:cluster:under_progress]]

Cluster Details

1.1 Total Resources

- **Compute nodes:** 7
- **Total slots:** 32
- **Architecture:** X86_64
- **Operating System:** Linux

1.2 Node Details

Node name	CPUs	RAM	GPUs	GPU Type	GPU	GPUs_Capability	Slots
		(GB ())			memory (GB ())		
hopper-1.coli.uni-saarland.de	128	1547	8	NVIDIA H100 PCIe	648	9.0	8
hopper-2.coli.uni-saarland.de	128	1031	8	NVIDIA H100 PCIe	648	9.0	8
hopper-3.coli.uni-saarland.de	128	1031	8	NVIDIA H100 PCIe	648	9.0	8
hopper-4.coli.uni-saarland.de	128	1030	8	NVIDIA H200 NVL	1145	9.0	8
tony-1.coli.uni-saarland.de	24	772	8	Tesla V100-PCIE-32GB	259	7.0	8
tony-2.coli.uni-saarland.de	24	191	8	Tesla V100-PCIE-32GB	259	7.0	8
tony-3.coli.uni-saarland.de	256	515	8	NVIDIA A100 80GB PCIe	649	8.0	8

Node name	CPUs	RAM	GPUs	GPU Type	GPU	GPUs_Capability	Slots
		(GB ())			memory (GB ())		
tony-4.coli.uni-saarland.de	256	515	8	NVIDIA A100-PCIE-40GB	323	8.0	8

2. Accessing the Cluster

- **SSH login:**

```
ssh user@login.lst.uni-saarland.de
```



- **Submit node login:**

```
ssh submit
```



3. Basic Workflow

1. Prepare your job script (''job.sub'')
2. Submit with ''condor_submit job.sub'' or ''condor_submit -i job.sub'' (for interactive sessions)
3. Check status with ''condor_q'' (can be running, idle, or held)
4. Retrieve output when finished



👉 See an Example of Job Script here.

4. Useful Commands

This section highlights the most essential commands and use cases. For the complete HTCondor commands reference, visit: [Full Documentation \(https://htcondor.readthedocs.io/en/latest/man-pages/index.html\)](https://htcondor.readthedocs.io/en/latest/man-pages/index.html).

Commands	Brief Explanation
condor_nodestate	Overview of free GPUs, CPUs, and memory
condor_q (https://htcondor.readthedocs.io/en/latest/tools/condor_q.html)	Show your current jobs
condor_q -allusers (https://htcondor.readthedocs.io/en/latest/tools/condor_q.html)	See all jobs in the system
condor_status (https://htcondor.readthedocs.io/en/latest/man-pages/condor_status.html)	Show all machine status
condor_submit (https://htcondor.readthedocs.io/en/latest/man-pages/condor_submit.html)	Submit a batch job
condor_submit -i (https://htcondor.readthedocs.io/en/latest/users-manual/submitting-a-job.html#interactive-jobs)	Submit an interactive job
condor_rm (https://htcondor.readthedocs.io/en/latest/man-pages/condor_rm.html)	Remove a job
condor_status -constraint 'RemoteUser == "[lst-email]"'	See which nodes are running your jobs

5. Storage Quota & File Areas

- **Home directory** (/nethome/\$USER) – Backed up, **100 GB () quota**
- **Store logs and projects** (/scratch/\$USER) – Not backed up
- **Check your usage:** quota -s

6. Job Scheduling

This policy ensures fair access, avoids resource starvation, and provides scheduling flexibility for all users.

Group	Target Users	GPU Limit	Preemptable?	Max Runtime	How to Use
Powerusers	PhDs and HiWis with PhD-level access	Up to 4 GPUs	 No	7 days	Default: submit normally – HTCondor auto-detects “poweruser” group.

Group	Target Users	GPU Limit	Preemptable?	Max Runtime	How to Use
Pausable	Flexible jobs by any group (if allowed)	Unlimited	✅ Yes	No hard limit	Add <code>accounting_group = pausable</code> to your <code>.sub</code> file to submit into the pausable queue.
Students	Undergrad and Master's students	1 GPU per default job	❌ (unless approved)	—	Default: 1 job at a time. To use pausable jobs: supervisor approval + <code>accounting_group = pausable</code>

⚠️ Important Notes:

- Powerusers Group:

1. Jobs are **non-preemptable** (cannot be interrupted).
2. To use more than 4 GPUs, submit additional jobs under the **pausable** queue.
3. If you need more than 4 GPUs at once, you can send an email request for this **via PI** to the systems group (help@lst). Once you finish your work, please send an email again to revert to the normal poweruser limits to ensure fairness.

- Pausable Group:

1. Jobs are **preemptable**: they may be interrupted to free resources for higher-priority jobs.
2. Jobs automatically **requeue** after being preempted.
3. ⚠️ Ensure your code is **checkpointable** or **resumable**; otherwise it will restart from scratch if interrupted.

- Students Group:

1. By default, students can submit only **one job** at a time.
2. Submitting pausable jobs requires **prior approval** from a supervisor.

7. Guidelines & Best Practices

- **Stage I/O locally** — Don't do frequent writes to `/scratch`` (because it is shared). First write to **local disk**, then copy/move to `/scratch`` whatever is needed.
- **Never work on the login/submit host** — Do not compile code, build environments, or run analyses on the head node. Use an interactive slot (``condor_submit -interactive``) or submit a dedicated job script for setup tasks.
- **Share common data** — Check `/scratch/common_models/`` before pulling large models or datasets. If missing, download once and place it there so others can benefit.
- **Be a good citizen** — Avoid flooding the queue with hundreds of tiny jobs; bundle short tasks or use job arrays.
- **When in doubt** — Talk to people inside the group when unsure.

- **Request only what you need** — In your submit file, explicitly set

```
request_cpus           # Number of CPU threads your job needs
request_memory         # Amount of RAM memory needed
request_gpus           # Number of GPUs required
+MaxRuntime            # Maximum runtime in seconds (24 hours for interac
tive jobs, 1 week for batch jobs)
GPUs_GlobalMemoryMb    # Minimum/maximum GPU memory (in MB) needed per GP
U
```



—this helps HTCondor place your jobs faster.

- **Limit interactive checks** — Avoid unnecessary interactive checks or (condor_ssh_to_job).
- **Consistent naming** — Use meaningful naming conventions for log files.
- **Proactive monitoring** — Regularly check job status and logs via HTCondor tools.

8. Troubleshooting

- **Stuck in queue?**

Check `condor_q -better-analyze <JOBID>`

- **Job on hold?**

Check `condor_q -hold <JOBID>`

Common causes: missing executables, permission issues, or exceeded quotas.

- **Storage full?**

Run ``

- **Unexpected job exits or errors?**

Inspect job log files (`.log` , `.out` , `err`)

9. Additional Resources

- User' manual (<https://htcondor.readthedocs.io/en/latest/users-manual/index.html#users-manual>)
- HT Condor tutorial (https://docs.google.com/presentation/d/1b04l6xBx8YKRAflyTvVRNW-dwMTJGm1rtetPEt9tZ9w/edit#slide=id.g34947be728d_0_217)
- Sample to tryout (https://drive.google.com/file/d/1wjwgC-a6m22Q52Rsw33ZhUb0KrajVbvX/view?usp=drive_link)

LST-Wiki
